

AUTHENTICLENS

KNOWLEDGE BASE - DOCUMENT 7 OF 9

Document 7 - Mobile Camera, Compression, Metadata, and Provenance Forensics

Phone Realism, Social Media Degradation, Screenshot Limits, and Trust Signals

Purpose: upload-ready operational documentation for the AuthenticLens Gemini Gem. This document translates theory into retrieval-friendly procedures, decision rules, prompt modules, and quality-control checks.

Prepared for: AuthenticLens - The Realism Architect

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Language: English (Canada)

Primary target: Gemini Pro + Nano Banana / Nano Banana Pro image generation and editing workflows

Purpose

Teach AuthenticLens how everyday phone cameras, social platforms, screenshots, metadata loss, and provenance systems affect image authenticity analysis and realism enhancement. This document reduces false AI accusations caused by normal digital imaging pipelines.

Intended Gemini Behaviour

- Recognize computational photography artifacts as normal in modern phone images.
- Separate compression damage from AI hallucination.
- Treat metadata absence as uncertainty, not proof of manipulation.
- Use provenance/Content Credentials when available, but do not assume availability.
- Prompt phone-realistic outputs differently from DSLR/film outputs.

A. Modern Phone-Camera Processing

Confirmed/operational guidance: smartphone cameras often combine small sensors, small lenses, automatic HDR, multi-frame stacking, tone mapping, denoising, sharpening, face/skin optimization, and computational depth effects. These processes can create images that look “too clean” without being AI-generated.

Artifact	Phone/computational cause	AI confusion risk	Classification instruction
Waxy skin	Noise reduction, face optimization, beauty filters, low light processing.	May look like AI skin smoothing.	Check anatomy, hairline, eyes, and texture transitions before increasing AI score.
HDR halos	Multi-frame HDR and local tone mapping near high-contrast edges.	May look like composite edges or AI glow.	If edges are globally consistent, classify as phone/HDR processing.
Over-sharpened outlines	Edge enhancement in phone processing or social apps.	May look like cutout/composite halo.	Look for uniform edge enhancement across entire image.
Flattened shadows	Shadow lift and HDR recovery.	May look physically impossible.	Compare with catchlights and contact shadows; phone HDR can reduce contrast.
Fake-looking depth blur	Computational portrait mode depth mask errors.	May look like AI bokeh.	Check whether depth errors follow segmentation boundaries rather than hallucinated geometry.
Colour shifts	Auto white balance under mixed light.	May look like synthetic colour grading.	Treat as edit/phone artifact unless scene logic fails.

B. JPEG and Social-Media Compression

Compression sign	Visual cue	Do not confuse with
Blocking	Square 8x8-like block structure in flat tones or shadows.	AI texture repetition.
Ringings/mosquito noise	Crawling dots or halos around text, edges, and high-contrast detail.	Pseudo-text or cutout halos.
Colour subsampling blur	Soft colour edges while luminance edges remain sharper.	Material melting.
Repeated recompression	Loss of fine texture, crunchy edges, uneven detail.	AI smoothing or fake detail.
Platform resizing	Uniform softness or aliasing after upload.	Model-generated low-detail background.

Operational rule: if a user uploads a screenshot or social-media download, increase uncertainty. Ask for the original camera file if authenticity matters. For creative enhancement, apply compression-aware correction gently; do not invent missing texture aggressively.

C. Screenshot and Repost Limitations

- Screenshots discard much original metadata and can introduce display scaling artifacts.
- Reposts can strip EXIF, colour profiles, and provenance information.
- A screenshot of a real photo is not the same forensic object as the original photo.
- Low-resolution reposts hide text detail, hand topology, and edge evidence; report uncertainty.

D. Metadata and EXIF: Useful but Not Decisive

Metadata signal	Use	Limitation
EXIF camera model	Can support a capture pipeline if plausible.	Can be stripped, edited, copied, or absent from exports.
Timestamp/location	Can support context if aligned with scene.	Can be wrong, spoofed, timezone-shifted, or removed.
Software tag	Can reveal editor/export tool.	Editor tag does not prove image is fake.
Colour profile	Helps web/product colour workflow.	Missing profile is common and not suspicious alone.
File history/provenance	Best available evidence when cryptographically bound.	Not universal; absence is not proof of AI.

E. C2PA / Content Credentials / Provenance

Confirmed source guidance: C2PA Content Credentials are designed to carry tamper-evident provenance data about how media was created or modified. A credential can include claims/assertions about creation, modification, tools, and AI use. Visual analysis should be treated separately from provenance verification.

Gemini instruction: “Do not treat visual style as provenance. If Content Credentials, EXIF, or original files are unavailable, state that the conclusion is visual/heuristic, not verified provenance.”

F. Phone-Realistic Prompt Modules

Goal	Prompt module
Natural phone photo	Make this look like a real modern smartphone photo: deep focus, realistic HDR tone mapping, mild edge enhancement, natural skin texture, slight compression, no dramatic DSLR bokeh.
Social upload realism	Apply only mild export-level compression and resizing artifacts. Keep faces and text readable. Do not add heavy grain or artificial damage.
Screenshot analysis	Treat this as a screenshot or repost. Increase uncertainty and focus on visible structural evidence only. Do not overclaim authenticity or AI generation.
Repair phone artifacts	Reduce excessive sharpening halos and denoise smearing while preserving natural phone-camera character. Restore detail cautiously without inventing pores or fabric.

Quality-Control Procedure

1. Identify whether file appears original, screenshot, repost, or compressed export.
2. Inspect compression artifacts before calling texture failures AI.
3. Check whether halos/noise/smoothing are uniform pipeline artifacts or localized manipulation.
4. Ask for original camera file only when needed for stronger authentication.
5. For enhancement, preserve a plausible capture pipeline: phone, DSLR, film, CCTV, or screenshot.

Quick Reference Summary

Phone artifacts are often global and pipeline-consistent; AI artifacts are usually structural or semantically incoherent. Compression can mimic many visual failures. Metadata helps but does not prove authenticity unless supported by reliable provenance.

Source Notes and Reliability Labels

Use source notes as operational grounding, not as a claim of legal or forensic certainty. If metadata, original files, or verified provenance are unavailable, AuthenticLens must label final authenticity claims as probabilistic.

- **Internal Doc 1:** AuthenticLens_Doc1_AI_Artifact_Lexicon.pdf - primary artifact taxonomy and detection guidance.

- **Internal Doc 2:** AuthenticLens_Doc2_Photographic_Physics.pdf - optical physics, sensor/film artifacts, lighting, exposure, and colour foundations.
- **Internal Doc 3:** AuthenticLens_Doc3_Realism_Prompting_Formulas.pdf - realism prompt formulas, camera/film language, and platform prompting references.
- **Google AI Developers:** Image generation with Gemini / Nano Banana and Nano Banana Pro documentation. Source note: confirms Gemini image generation/editing is conversational, supports text+image inputs, identifies Nano Banana as gemini-2.5-flash-image and Nano Banana Pro as gemini-3-pro-image-preview, and states generated images include SynthID watermarking.
- **Adobe Lightroom Help:** Lightroom documentation for Texture, Clarity, Dehaze, Vignette, Grain, Sharpening, Noise Reduction, local adjustments, and masking. Source note: used to separate editing artifacts from AI-generation artifacts.
- **C2PA Specification:** C2PA Content Credentials specification and explainer. Source note: provenance is represented by signed manifests / Content Credentials that can record creation and modification assertions.
- **NIST Digital Media / Deepfake Guidance:** NIST digital and multimedia evidence resources and deepfake guidance. Source note: provenance, supply chain, compression, and context matter; visual inspection alone is not definitive.
- **W3C sRGB:** sRGB as the default colour space for web images. Source note: used for e-commerce and social-media colour delivery guidance.

Reliability labels used in this knowledge base: Confirmed source, Common photography practice, Image-forensics heuristic, Operational guidance, Prompting heuristic, Model-behaviour observation.